Homework 2

Complete all 10 questions on probability distributions, simulation, estimation, and one-sample inference.

Submission instructions

Submit one PDF or Word document containing all answers, derivations, code, and code output. Clearly label each question and subpart. Scans of handwritten work and screenshots of code, output, and figures are allowed; they must be legible and included in the same file. Do not submit code or output only as separate files or external links.

For Questions 2–6, show the hand derivations and arithmetic; a calculator is allowed, but software output does not replace the work. Questions 7 and 9 require Python; Questions 8 and 10 require R. Include interpretations as well as numerical results. Report at least four decimal places where useful, retain precision until the last step, and use scientific notation for very small probabilities. All example data are illustrative.

Question 1 Interpreting statistical statements

Short answer

Answer each part in two to four sentences. Use precise language about what is random and what is fixed.

(a) Distinguish a population parameter, an estimator, and an estimate. Give one example of each when estimating a population mean.

(b) Explain the difference between the standard deviation of individual observations and the standard error of a sample mean. Under independent, identically distributed sampling with finite variance, what happens to the standard error when the sample size is multiplied by four?

(c) Correct this statement: “A 95% confidence interval contains 95% of individual observations, and there is a 95% frequentist probability that the fixed population mean lies inside the particular interval already calculated.”

(d) Correct this statement: “A p-value of 0.03 means that the null hypothesis has a 3% probability of being true; a p-value above 0.05 proves the null hypothesis.”

Question 2 A discrete probability distribution

Hand calculation

The number of positive responses in a group of three is modeled by the following probability mass function. Do not assume a binomial model.

x	0	1	2	3
P(X = x)	0.10	0.20	0.40	0.30

(a) Verify that the table is a valid probability mass function. Write the complete piecewise cumulative distribution function $F_X(t) = P(X \le t)$, including values outside the support.

(b) Calculate $F_X(1.5)$, $P(X \ge 2)$, and $P(X > 2)$. Explain why the last two answers differ.

(c) Calculate $E(X)$, $E(X^2)$, $Var(X)$, and $SD(X)$ directly from the table. Explain how the mean can be a value that the random variable cannot take.

Question 3 Binomial counts and Poisson counts

Hand calculation

Use the assumptions stated in each scenario. Show the probability formulas and substitutions.

(a) Eight independent patients each have response probability 0.25. Let X count responses. State the distribution and the four binomial conditions. Calculate $P(X = 2)$ and $P(X \geq 1)$.

(b) Write X as a sum of Bernoulli indicators. Derive its mean and variance, explaining where independence is used.

(c) Calls arrive according to a homogeneous Poisson process at a rate of 6 per hour. Let Y count calls in 45 minutes. State its distribution and calculate $P(Y = 0)$, $P(Y \geq 2)$, $E(Y)$, and $Var(Y)$.

(d) Explain why the parameter for 45 minutes is not 6. Give one feature of real call arrivals that could make the homogeneous Poisson model inappropriate.

Question 4 Densities and continuous probabilities

Hand calculation

A continuous random variable has the density below, where c is an unknown constant.

$$f_X(x) = \begin{cases} cx^2, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Determine c by integration, and write the complete piecewise CDF.

(b) Calculate $P(0.25 < X < 0.75)$, $E(X)$, $E(X^2)$, and $Var(X)$ using integrals.

(c) Calculate $f_X(0.8)$ and $P(X = 0.8)$. Explain why a density value greater than one is valid.

(d) Separately, let $W \sim Uniform(4,16)$. Calculate $P(7 < W < 13)$, $E(W)$, and $Var(W)$.

Question 5 Normal measurements and sample means

Hand calculation

For parts (a) and (b), suppose independent measurements follow $N(100, 12^2)$, where 12 is the population standard deviation.

(a) Find the probability that one measurement exceeds 118, and find the 90th percentile of individual measurements. Show the standardization or quantile equation.

(b) For a sample of 36 measurements, derive $E(\overline{X})$, $Var(\overline{X})$, and $SE(\overline{X})$. State the exact sampling distribution of $\overline{X}$ and calculate $P(\overline{X} > 104)$.

(c) Now treat the population mean μ as unknown while the population SD remains known to be 12. A sample of size 36 has mean 103. Derive and calculate a two-sided 95% confidence interval for μ.

(d) With the same known population SD, find the smallest planned sample size giving a 95% confidence interval with margin of error at most 3. Show the inequality and round in the correct direction.

(e) If the population is not normal but observations remain iid with finite, positive variance, which moment and SE results remain exact? What does the central limit theorem imply about the sampling distribution? Explain why a universal rule of "n at least 30" is not a guarantee.

For hand calculations, use the supplied values: $\Phi(1.5) = 0.933193$, $\Phi(2) = 0.977250$, $z_{0.90} = 1.281552$, and $z_{0.975} = 1.959964$. All quantiles in this homework use left-tail probabilities.

Question 6 Inference with an unknown population variance

Hand calculation

The following nine illustrative measurements are an independent sample from a normal population with unknown mean and variance. The same values are supplied in measurements.csv.

$$94,\ 96,\ 98,\ 100,\ 100,\ 100,\ 102,\ 104,\ 106$$

(a) Calculate n, $\overline{x}$, the sum of squared deviations, the unbiased sample variance s^2, the sample SD s, and the estimated standard error of the mean. Show the arithmetic.

(b) State the t pivot and derive a two-sided 95% confidence interval for the population mean. Explain why the reference distribution has $n - 1$ degrees of freedom and why 1.96 is not the correct critical value here.

(c) Test $H_0: \mu = 96$ against $H_1: \mu \neq 96$ at significance level 0.05. Calculate the test statistic, state the rejection rule, write the two-sided p-value expression, and give the decision and a contextual conclusion. Verify agreement with part (b). A numerical p-value is optional; a correct CDF expression is sufficient.

(d) State the chi-square pivot, invert its central 95% probability statement, and calculate confidence intervals for the population variance and population SD. Explain why the larger chi-square quantile is used in the lower endpoint and state the normality assumption.

Use $t_{8,0.975} = 2.306004$, $\chi^2_{8,0.025} = 2.179731$, and $\chi^2_{8,0.975} = 17.534546$.

Question 7 Exact probabilities and approximations in Python

Python coding

Use NumPy and SciPy. Set numpy.random.default_rng(207) and use 100,000 independent draws for each simulation. Process cases A, B, C, and then the rare-event example in that order.

Case	Model and target	Approximation to evaluate
A	X ~ Binomial(200, 0.01); P(X ≥ 4)	Poisson with mean np
B	X ~ Binomial(100, 0.30); P(X ≥ 40)	Normal with continuity correction
C	X ~ Poisson(25); P(X ≥ 35)	Normal with continuity correction

(a) For each case, calculate the exact probability, the stated approximation, absolute error, and relative error. Define relative error as absolute error divided by the exact probability. State every approximation parameter and the corrected normal cutoff.

(b) Simulate each count model and estimate the same probability. Present a table comparing exact, approximate, and simulated values. Explain whether the approximation is useful for each event, distinguishing absolute from relative error.

(c) For $R \sim Binomial(4000, 0.0001)$, calculate $P(R \geq 10)$ accurately with a survival function. Simulate 100,000 draws, report the number satisfying the event, and calculate the expected number of such occurrences. Explain why observing zero does not establish that the probability is zero.

Include your Python code, the numerical comparison table, and the rare-event output. Use binom.sf(k - 1, ...) or poisson.sf(k - 1, ...) for an inclusive upper tail.

Question 8 Inverse transform simulation in R

R coding

Use only runif for random-number generation; qnorm is allowed as a deterministic quantile function. Set set.seed(208), use 100,000 draws per target, and generate three separate uniform vectors in the order Normal, Bernoulli, polynomial.

(a) State the generalized inverse $F^{-1}(u) = inf\{x : F(x) \geq u\}$ for $0 < u < 1$. Derive transformations from $U \sim Uniform(0,1)$ to $Y \sim N(50, 8^2)$, $D \sim Bernoulli(0.30)$, and a variable X with the density in Question 4. Use the generalized-inverse convention for the Bernoulli threshold.

(b) Implement the three transformations. Report sample means and sample variances alongside their theoretical values. Estimate $P(Y > 58)$, $P(D = 1)$, and $P(X \leq 0.5)$ and compare with theory.

(c) Make density-scale histograms for the Normal and polynomial variables, overlaying their theoretical density curves. Explain why the two continuous transformations give the specified CDFs.

(d) State the continuous probability integral transform. For the Bernoulli variable, list the possible values and probabilities of $F_D(D)$ and explain why it is not uniform. Explain why reusing one uniform vector for all three transformations would not produce independent columns.

Include the R code, output tables, both plots, and explanations. Do not call rnorm, rbinom, or a direct sampler for the polynomial variable.

Question 9 Constructing t and chi-square distributions in Python

Python coding

Use 100,000 replications and numpy.random.default_rng(209). For each degree of freedom, generate the normal matrix first and a separate normal numerator second; process 5 df before 30 df.

(a) For $\nu = 5$ and $\nu = 30$, generate independent standard normal values and construct $Q = \sum_{j=1}^{\nu} Z_j^2$ and $T = Z_0/\sqrt{Q/\nu}$, with Z_0 independent of all terms in Q. Do not use direct chi-square or t random samplers. Explain why independence matters.

(b) Report the sample mean and variance of Q and T, and compare them with their theoretical values. Estimate $P(Q > \chi^2_{\nu,0.95})$ and $P(|T| > 1.96)$; compare with the correct theoretical tail probabilities. Explain the change in t tails as the degrees of freedom increase and state when the t mean and variance exist.

(c) For 5 df, plot density-scale histograms of Q and T with their theoretical densities. Add the standard normal density to the t panel.

(d) In a separate sample of 100 independent observations, four categories have observed counts (18, 22, 26, 34). Under the null, each category has fixed probability 0.25, with no parameters estimated from these counts. Calculate the Pearson goodness-of-fit statistic, degrees of freedom, and approximate p-value. State the decision at 0.05 and the conditions supporting the chi-square reference.

Include code, output tables, both plots, and interpretations. Use sample variance with ddof=1.

Question 10 Coverage and testing in R

R coding

Simulate independent normal samples with mean 100 and SD 12. Set set.seed(210), use 20,000 replications, and process sample sizes 10 and then 40. For each size, use matrix(rnorm(B*n, 100, 12), nrow=B), with one sample per row.

(a) For each sample, construct three nominal 95% mean intervals: a t interval using the sample SD, a z interval using the known SD of 12, and a z interval incorrectly substituting the sample SD for the known SD. Estimate each method's coverage of the true mean. Also report the average width of the t interval.

(b) For the same samples, test $H_0: \mu = 100$ against $H_1: \mu \neq 100$ using a two-sided t test at 0.05. Estimate the Type I error rate. For every replication, check whether rejection agrees with exclusion of 100 from the t interval, and report the number of disagreements.

(c) Add 4 to every observation in the same samples, so the true mean becomes 104. Apply the same t test against 100. Estimate power and Type II error for each sample size.

(d) Plot the three coverage estimates by sample size, with a horizontal line at 0.95. Interpret coverage, width, Type I error, and power. Explain why the interval using z with the sample SD undercovers, particularly for the smaller sample.

Include code, the numerical results, the coverage plot, and explanations. Identify what is counted in each simulated proportion; do not describe power as the probability that the alternative hypothesis is true.